

Notes: Rangan (JFE, 1998)
Earnings Management and the Performance of Seasoned Equity Offerings

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1 Big picture

This paper studies whether the poor post-issue performance of firms conducting seasoned equity offerings (SEOs) can be partly explained by **earnings management around the offering**. The key idea is straightforward: if issuing firms use discretionary accruals to temporarily raise reported earnings, and investors fail to fully recognize that these accruals are transitory, the market will temporarily overvalue the issuing firms. When the accruals reverse and earnings fall after the offering, stock prices adjust downward.

Core message. SEO firms appear to manage earnings upward around the offering date, especially in the offering-announcement quarter and the following quarter. These discretionary accruals predict both subsequent earnings declines and subsequent market-adjusted stock underperformance. This supports the interpretation that the market does not fully undo the temporary earnings component created by accruals.

2 Incremental contribution of the paper

Main incremental contribution. The paper links three objects in one empirical design: **earnings management around SEOs**, **post-offering earnings declines**, and **post-offering abnormal stock returns**. It shows that discretionary accruals around the offering are not only high, but also have predictive power for the later earnings and return reversals.

More specifically, the paper contributes in five ways.

1. **Focuses on the year around the offering.** Unlike some related work that studies accruals before the offering or long-run returns over several years, Rangan examines discretionary accruals during the year surrounding the offering and predicts returns in the next year, when the accrual reversal is most direct.
2. **Uses quarterly timing.** The paper identifies exactly when accruals rise: quarter 0, the quarter containing the first earnings announcement after the offering announcement, and quarter 1, the next quarter.
3. **Separates timing from manipulation.** The paper recognizes that firms might simply time offerings after high accrual quarters. It estimates a joint system to show that high quarter-0 accruals are not only a timing effect; issuing status is itself positively related to current accruals.
4. **Connects accruals to subsequent operating performance.** Discretionary accruals in year 0 predict earnings declines in year 1, consistent with reversal of temporary accrual-based earnings increases.
5. **Connects accruals to market inefficiency.** Year-0 discretionary accruals predict year-1 market-adjusted returns, even after controlling for sales growth, capital expenditure growth, firm size, and book-to-market.

Why this is new. The paper moves beyond documenting that SEO firms underperform. It provides a mechanism: managers appear to boost earnings through accruals near the offering, and the stock market appears to price the managed earnings too optimistically.

3 Economic mechanism

The mechanism has three steps.

1. **Incentive.** Current shareholders of issuing firms benefit if the firm can sell equity at a high valuation. This gives managers an incentive to report temporarily high earnings near the offering.
2. **Tool.** Managers can use current accruals, such as credit sales, receivables, inventories, and current liabilities, to shift income across periods without necessarily violating accounting rules.
3. **Market reaction.** If investors treat the accrual-driven earnings increase as persistent rather than temporary, the firm is overvalued at issuance. When accruals reverse, earnings fall and stock prices decline.

The argument therefore combines corporate financing incentives, accounting discretion, and market mispricing. The paper does not need all SEO underperformance to be explained by accruals; it asks whether accruals explain a significant and economically meaningful part of the phenomenon.

4 Data and sample

The starting sample consists of registered firm-commitment SEOs by U.S. firms from 1987 through 1990. The final sample contains **230 offerings**: 159 primary offerings, 49 mixed primary-secondary offerings, and 22 pure secondary offerings.

The author excludes financial firms, shelf offerings, combined debt-equity offerings, offerings with warrants, firms lacking Compustat data, repeat offerings by the same firm during the sample period, and firms without enough data to estimate discretionary accruals.

The average offering is economically meaningful: the mean value of shares offered is about \$53 million, and the mean ratio of shares offered to pre-offering shares outstanding is 0.23. The average market value of equity before the offering announcement is about \$650 million.

5 Empirical design

5.1 Event-time convention

The event clock is organized around the offering announcement and nearby earnings announcements. Quarter -1 is the quarter with the last earnings announcement before the offering announcement. Quarter 0 is the quarter with the first earnings announcement after the offering announcement. Quarters 1 and 2 follow sequentially.

The author predicts that earnings management should be most likely from quarter -1 through quarter 2, because managers want high reported earnings before the offering and may also want to avoid an immediate post-offering earnings reversal.

5.2 Measuring discretionary accruals

The paper uses current accruals:

$$ACC = (\Delta CA - \Delta CASH) - (\Delta CL - \Delta STD),$$

where current accruals are the change in noncash current assets minus the change in current liabilities excluding the current portion of long-term debt.

The nondiscretionary component is estimated from a firm-specific pre-event regression:

$$ACC_{it} = \beta_{0i} + \beta_{1i}\Delta REV_{it} + \beta_{2i}\Delta COGS_{it} + \varepsilon_{it}.$$

Following the modified Jones-style logic, predicted accruals in the event period adjust revenue changes by changes in accounts receivable:

$$DISC_{ip} = ACC_{ip} - (b_{0i} + b_{1i}[\Delta REV_{ip} - \Delta REC_{ip}] + b_{2i}\Delta COGS_{ip}).$$

Discretionary accruals are scaled by beginning assets. The main annual accrual variable, $DISC_0$, is the sum of discretionary accruals over quarters -1 through 2 , scaled by assets at the beginning of the four-quarter period.

5.3 Earnings and return tests

The earnings test asks whether year-0 discretionary accruals predict changes in return on assets in years 1 to 3:

$$\Delta ROA_j = a_0 + a_1 DISC_0 + \varepsilon_j, \quad j = 1, 2, 3.$$

The return test asks whether the stock market understands the future reversal implied by $DISC_0$. If the market fully prices the future earnings decline, $DISC_0$ should not predict subsequent abnormal returns after controlling for unexpected year-1 earnings. If the market is surprised by the reversal, $DISC_0$ should negatively predict year-1 abnormal returns.

6 Main findings

1. **Discretionary accruals spike around the offering.** Median discretionary accruals are 0.83 percent of assets in quarter 0 and 1.15 percent in quarter 1. These are statistically significant and economically large relative to median ROA.
2. **The spike is not only timing.** The paper finds evidence that firms do time offerings after high accruals, but the joint estimation also indicates active earnings management in quarter 0.
3. **Earnings decline after the offering.** Median change in ROA is positive before and during year 0, then turns negative in the following years. In year 1, median ΔROA is -0.97 percent, and 64 percent of issuers have negative earnings changes.
4. **Discretionary accruals predict year-1 earnings declines.** In Table 3, the coefficient of $DISC_0$ in the year-1 earnings-change regression is -0.155 with a t-statistic of -4.85 . The relation is concentrated in year 1, consistent with reversal of accrual-based earnings management.

5. **SEO firms underperform after the offering.** Table 4 reports a median year-1 market-adjusted return of -7.38 percent and a median year-1 earnings-announcement return of -1.32 percent.
6. **Discretionary accruals predict year-1 stock returns.** The coefficient of $DISC_0$ in the full-year abnormal return regression is about -0.55 , with a t-statistic around -4.07 . A one-standard-deviation increase in year-0 discretionary accruals is associated with roughly a 10 percent decline in year-1 market-adjusted returns.

Bottom line. The results support a market-overvaluation interpretation: investors appear not to fully understand that accrual-driven earnings growth around SEOs is temporary.

7 Interpretation

The paper’s interpretation is that issuing firms benefit from temporary misvaluation. Managers raise reported earnings using discretionary accruals, the market does not completely discount the earnings boost, and the firm issues equity while overvalued. When accruals reverse, earnings fall and investors revise valuations downward.

The result is closely related to Sloan’s accrual anomaly, but with a corporate finance event that creates a clear managerial incentive. SEOs provide a setting in which the benefits of higher reported earnings are especially direct: issuing firms can sell equity at more favorable terms.

8 Identification and limitations

Strengths.

- The quarterly event-time design helps locate the timing of accruals relative to the offering.
- The paper directly connects accruals to both future operating performance and future stock returns.
- The joint issue-accrual model helps address the concern that firms merely time offerings after naturally high accrual quarters.
- The stock-return tests control for other known predictors, including sales growth, capital expenditure growth, firm size, and book-to-market.

Limitations.

- Discretionary accruals are noisy proxies for intentional earnings management.
- The sample period is relatively short and contains 230 offerings, so external validity should be interpreted carefully.
- The paper does not claim that accruals explain all long-run SEO underperformance; the strongest evidence is for the year after the offering.
- The analysis is not randomized. Offering timing, firm fundamentals, and managerial reporting choices remain jointly determined.

Best interpretation. The paper provides evidence that earnings management is one channel through which SEO firms can be temporarily overvalued, and that the market corrects this overvaluation when accruals reverse and earnings disappoint.

9 Extracted figure and tables

9.1 Table 1 and Figure 1: sample and event timing

Table 1
Descriptive statistics for seasoned equity offerings

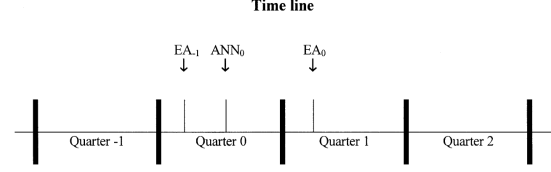
The summary statistics are based on a sample of 230 registered firm-commitment common stock offerings by US firms for the years 1987 through 1990. The sample consists of 159 primary offerings, 49 offerings with both a primary and a secondary component, and 22 pure secondary offerings. Offering information is from the Directory of Corporate Financing and the 1993 Quarterly Primary, Supplemental, Tertiary, and Full Coverage Compustat files.

	Mean	Median	Std. Dev.	First quartile	Third quartile
Market value of equity ^a (in millions of \$)	649.62	180.13	1583.00	66.17	538.86
Value of shares offered (in millions of \$):					
All offerings	53.17	30.50	88.30	15.00	59.50
Secondary component ^b	34.53	9.42	76.35	3.06	36.75
Shares offered/pre-offering shares outstanding:					
All offerings	0.23	0.16	0.29	0.09	0.27
Secondary component	0.13	0.08	0.27	0.04	0.13
Offering size/market value of equity:					
All offerings	0.25	0.16	0.31	0.09	0.30
Secondary component	0.13	0.07	0.28	0.03	0.14
Pre-offering leverage ^c	0.25	0.25	0.16	0.11	0.35

^aMarket value of equity is measured at the end of the quarter before the offering announcement date.

^bStatistics on the secondary component are based on the sample of mixed offerings and pure secondary offerings.

^cPre-offering leverage is the ratio of book value of long term debt to total assets at the end of the quarter before the offering announcement date.



EA_{-1} = Last earnings-announcement date before the offering announcement.

EA_0 = First earnings-announcement date after the offering announcement.

ANN_0 = Offering Announcement date.

Fig. 1. Time line, EA_{-1} = last earnings-announcement date before the offering announcement; EA_0 = first earnings-announcement date after the offering announcement; ANN_0 = offering announcement date.

Figure 1. Event-time convention around the offering announcement.

Table 1. Descriptive statistics for seasoned equity offerings.

Read:

- Table 1 shows that the sample contains economically important equity issues: the mean value of shares offered is \$53.17 million, and the mean number of shares offered equals 23 percent of pre-offering shares outstanding.
- Figure 1 defines the event clock. Quarter -1 ends with the last earnings announcement before the offering announcement; quarter 0 contains the first earnings announcement after the offering announcement.
- This timing matters because the predicted earnings-management window is centered on the offering announcement and nearby earnings announcements.

9.2 Tables 2 and 3: discretionary accruals and subsequent earnings

Table 2
Median discretionary accruals around seasoned equity offerings

The table reports median discretionary accruals and return on assets around seasoned offerings for a sample of 230 firms. Discretionary accruals are the difference between realized accruals in quarters -4 to 3 and predicted accruals in those quarters. Predicted accruals for each firm are forecasts based on the following regression:

$$ACC_{it} = \beta_{0i} + \beta_{1i}\Delta REV_{it} + \beta_{2i}\Delta COGS_{it} + e_{it}$$

where

ACC_{it} = current accruals for firm i in quarter t ;
 ΔREV_{it} = change in revenue for firm i in quarter t ;
 $\Delta COGS_{it}$ = change in cost-of-goods-sold expense for firm i in quarter t .

The estimation period for the above regression ends in quarter -5 . To enter the sample, a firm is required to have at least six consecutive estimation period observations. The sample estimation period length averages 14 and ranges from 6 to 27. Discretionary accruals for each quarter p in the event period (quarter -4 to quarter 3) is defined as:

$$DISC_{ip} = ACC_{ip} - (b_{0i} + b_{1i}[\Delta REV_{ip} - \Delta REC_{ip}] + b_{2i}[\Delta COGS_{ip}]),$$

where b_{0i} , b_{1i} , and b_{2i} are the estimated coefficients from the first stage regression and ΔREC_{ip} is the change in accounts receivable for firm i in quarter p . Discretionary accruals are scaled by beginning assets. Return on assets is income before extraordinary items scaled by beginning assets.

Quarter	Median DISC	Wilcoxon p -values	% positive	Median ROA
-4	0.13%	0.43	51%	1.37%
-3	0.17%	0.37	52%	1.31%
-2	0.19%	0.29	54%	1.44%
-1	0.10%	0.15	52%	1.60%
0	0.83%	0.02	57%	1.74%
1	1.15%	0.00	61%	1.67%
2	0.15%	0.29	52%	1.54%
3	0.52%	0.13	54%	1.53%

Table 2. Median discretionary accruals around SEOs.

Table 3
Discretionary accruals in year 0 and subsequent earnings changes

Panel A reports median changes in return on assets (ΔROA) for years 1 to 3. ROA for year 0 is income before extraordinary items summed over quarter -1 to quarter 2 and scaled by assets at beginning of four-quarter period. Performance over four quarter intervals prior to and subsequent to year 0 are similarly aggregated to construct ROA for years -2 to 3. ΔROA are computed as first differences in annual ROA .

Panel B reports results from cross-sectional regressions of ΔROA in years 1 to 3 on year 0 variables. The first three columns contain results from three univariate regressions. The dependent variables in these regressions are ΔROA in years 1, 2, and 3. The independent variable in each regression is year 0 discretionary accruals ($DISC_0$). $DISC_0$ is discretionary accruals summed over quarter -1 to quarter 2 and scaled by assets at the beginning of the four quarter period. The last three columns of panel B report results from multiple regressions of ΔROA in year 1 on $DISC_0$, year 0 sales growth ($SGRO_0$), and year 0 capital expenditure growth ($CAPEX_0$). $SGRO_0$ is computed as percentage growth rate in sales from year -1 to year 0. $CAPEX_0$ is computed as change in capital expenditures from year -1 to year 0 scaled by assets at the end of year -1 . All regressions exclude observations with values of $DISC_0$ or ΔROA in excess of 100% in absolute value. The numbers in parentheses are t -statistics.

Panel A: Median change in ROA

Year	-2	-1	0	1	2	3
Median ΔROA	0.21%	0.33%	0.19%	-0.97%	-0.67%	-0.23%
Wilcoxon p -value	0.22	0.00	0.01	0.00	0.00	0.07
% negative	48%	44%	46%	64%	60%	53%
# of obs.	212	228	229	230	230	229

Panel B: Regressions of ΔROA in year 1 to 3 on year 0 variables

Independent variable	Dependent variable					
	ΔROA_1	ΔROA_2	ΔROA_3	ΔROA_1	ΔROA_1	ΔROA_1
$DISC_0$	-0.155 (-4.85) ^a	-0.006 (-0.20)	-0.009 (-0.29)	-0.101 (-3.41) ^a	-0.140 (-3.94) ^a	-0.103 (-2.92) ^a
$SGRO_0$				-0.093 (-7.29) ^a		-0.078 (-4.00) ^a
$CAPGRO_0$					-0.128 (-3.62) ^a	-0.091 (-2.58) ^a
# of obs.	227	229	217	227	178	178
Adjusted R^2	9.05%	-0.42%	-0.41%	26.17%	14.33%	21.08%

^aSignificant at the 1% level based on a two-sided test.

Table 3. Discretionary accruals in year 0 and subsequent earnings changes.

Read:

- Table 2 is the first key result. Median discretionary accruals are small in most quarters but jump to 0.83 percent in quarter 0 and 1.15 percent in quarter 1.
- Table 3 shows that earnings rise into the offering and decline afterward. Median ΔROA is -0.97 percent in year 1, and 64 percent of issuers have negative earnings changes.
- The regression panel shows that $DISC_0$ strongly predicts the year-1 earnings decline, but not year-2 or year-3 changes. This supports a short-run reversal mechanism rather than a persistent relation.

9.3 Table 4: discretionary accruals and stock returns

Table 4

Discretionary accruals in year 0 and stock returns in year 1

Abnormal returns are computed by subtracting the return on the value-weighted market index from firm returns. Annual returns for year 1, which consists of quarters 3 to 6, are computed from the day after the earnings announcement for quarter 2 to the day of the earnings announcement for quarter 6. Year 1 earnings-announcement returns are compounded over the four earnings-announcement periods in the year. Earnings-announcement periods consist of the day preceding and the day of the Compustat earnings-announcement date. All returns are compounded daily.

Panel A reports median abnormal returns for the entire year and for the earnings-announcement periods. Panel B reports regressions of year 1 abnormal returns on year 0 variables and UE_1 , the unexpected component of year 1 earnings changes. The year 0 variables are discretionary accruals ($DISC_0$), sales growth ($SGRO_0$), and capital expenditure growth ($CAPGRO_0$). UE_1 is the residual from a regression of year 1 earnings changes (ΔROA_1) on year 0 variables. All regressions include log of firm size and book-to-market ratio at the end of the fiscal year before the offering as control variables. Regressions exclude four observations with year 1 abnormal returns greater than 150% and three observations with ΔROA_1 or $DISC_0$ greater than 100% in absolute value. The numbers in parentheses are t -statistics.

Panel A: Median year 1 abnormal returns

	One-year returns	Earnings-announcement returns
Median	− 7.38%	− 1.32%
Wilcoxon p -value	0.00	0.01
% negative	61%	59%
# of obs.	230	230

Panel B: Regressions of year 1 market-adjusted returns on year 0 variables

Independent variable	Dependent variable					
	One-year returns			Earnings-announcement returns		
UE_1	1.605 (5.84) ^a	1.883 (6.26) ^a	1.708 (4.98) ^a	0.227 (2.81) ^a	0.333 (3.70) ^a	0.329 (3.20) ^a
$DISC_0$	− 0.550 (− 4.07) ^a	− 0.518 (− 3.79) ^a	− 0.610 (− 3.74) ^a	− 0.054 (− 1.37)	− 0.062 (− 1.52)	− 0.092 (− 1.88) ^c
$SGRO_0$		− 0.034 (− 0.58)	0.082 (0.94)		0.021 (1.19)	0.062 (2.38) ^b
$CAPGRO_0$			− 0.299 (− 1.88) ^c			− 0.101 (− 2.12) ^b
# of obs.	223	223	174	223	223	174
Adjusted R^2	18.87%	20.23%	19.24%	6.35%	8.89%	11.56%

^aSignificant at the 1% level based on a two-sided test.

^bSignificant at the 5% level based on a two-sided test.

^cSignificant at the 10% level based on a two-sided test.

Table 4. Discretionary accruals in year 0 and stock returns in year 1.

Read:

- Panel A documents poor year-1 performance: the median one-year market-adjusted return is −7.38 percent, and the median earnings-announcement return is −1.32 percent.
- Panel B shows that $DISC_0$ predicts year-1 returns. The full-year abnormal return coefficient is about −0.55 in the simplest specification and remains negative after adding sales growth and capital expenditure growth.
- Economically, a one-standard-deviation increase in discretionary accruals is associated with roughly 10 percentage points lower year-1 market-adjusted returns.

10 Takeaway for future research

For studies of security issuance, this paper is a warning that reported accounting performance around the issue may not be a clean signal of fundamentals. Issuance events can create strong incentives for earnings management, and market reactions may not fully adjust for the transitory component of accruals.

For studies of market efficiency, the paper provides an event-based version of the accrual anomaly: investors appear to overvalue accrual-driven earnings, and this mispricing is especially relevant when firms are raising equity.

For empirical work, the paper motivates using both accounting information and issuance timing to study why equity issuers underperform after offerings.